INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

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Project Proposal on Smart Automatic Darkness Detection Lighting System

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INTRODUCTION :

In modern life, efficient use of electrical energy is very important. However, manual control of lighting systems often leads to unnecessary energy wastage, as lights may remain ON during daytime or OFF during darkness.

This project presents an **Automatic Darkness Detection and Lighting Control System** that uses a Light Dependent Resistor (LDR) to detect ambient light intensity. The system automatically turns the light ON in darkness and OFF in the presence of sufficient light, eliminating the need for manual operation.The system is simple, cost-effective, and suitable for applications such as street lighting, home automation, and garden lighting, helping to reduce energy consumption and improve convenience.

PROBLEM STATEMENT :

In everyday life, lighting systems are mostly controlled manually, which often leads to inefficient use of electrical energy. Lights are frequently left ON during daylight hours or are not switched ON at the right time during darkness due to human negligence or inconvenience. This results in unnecessary energy consumption, increased electricity costs, and reduced system efficiency.
In public places such as streets, parking areas, and gardens, manual control of lighting becomes even more impractical, as it requires continuous human involvement. Additionally, in many cases, there is no intelligent mechanism to adjust lighting based on surrounding light conditions, leading to poor energy management.Therefore, there is a need for a simple, reliable, and cost-effective automated lighting system that can detect ambient light levels and control the switching of lights accordingly. Such a system would help minimize energy wastage, reduce dependency on manual operation, and improve overall efficiency and convenience.

OBJECTIVES :

The main objectives of this project are :

- To design and develop an automatic lighting system based on ambient light conditions
- To reduce electrical energy wastage by eliminating manual switching
- To implement a low-cost and efficient system using an LDR sensor
- To ensure automatic ON/OFF control of lights depending on surrounding light intensity
- To improve user convenience through a fully automated operation
- To enhance system reliability and energy efficiency
- To apply basic concepts of sensor-based automation in practical applications
- To develop a system suitable for real-life uses such as street lighting, home automation, and outdoor lighting systems.

METHODOLOGY :

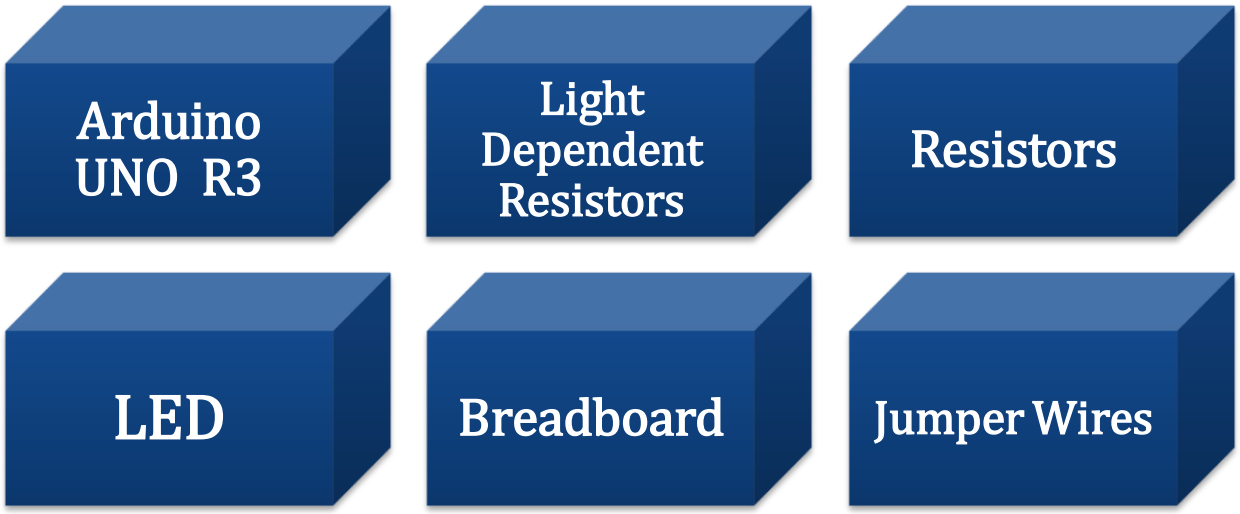
The proposed system is designed to automatically control lighting based on the surrounding light intensity using a Light Dependent Resistor (LDR). The LDR is used as the primary sensing element, which continuously monitors the ambient light level and changes its resistance accordingly.

In bright conditions, the resistance of the LDR decreases, while in darkness, its resistance increases. This variation in resistance is used to generate a corresponding voltage signal through a voltage divider circuit. The output voltage is then fed into a transistor or comparator circuit, which acts as a switching mechanism.When the ambient light intensity falls below a predefined threshold, indicating darkness, the circuit activates the switching component (such as a transistor or relay), turning the light ON. Similarly, when sufficient light is detected, the circuit deactivates the switch, turning the light OFF.

The system is powered by a suitable power supply and can be implemented using basic electronic components, making it simple, reliable, and cost-effective. Proper calibration of the threshold level ensures accurate operation under different environmental conditions.

COMPONENTS REQUIRED :

Hardware Components:



Arduino UNO R3	• Used as the main controller for processing input and controlloing output.
LDR	• Light Dependent Resistor is used to sense ambient light intensity
Resistor	• Used for voltage division and proper conditioning.
LED	• Used as the output lighting element.
Bread-board	• Used for circuit prototyping and connections.
Jumper Wires	• Used to establish electrical connections between components.

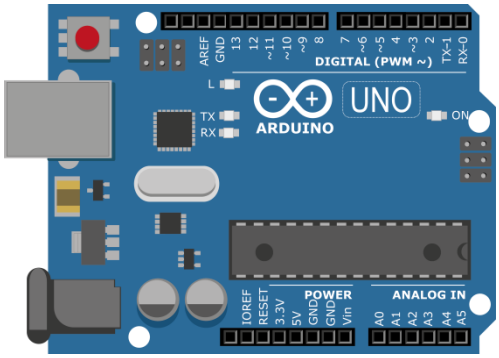
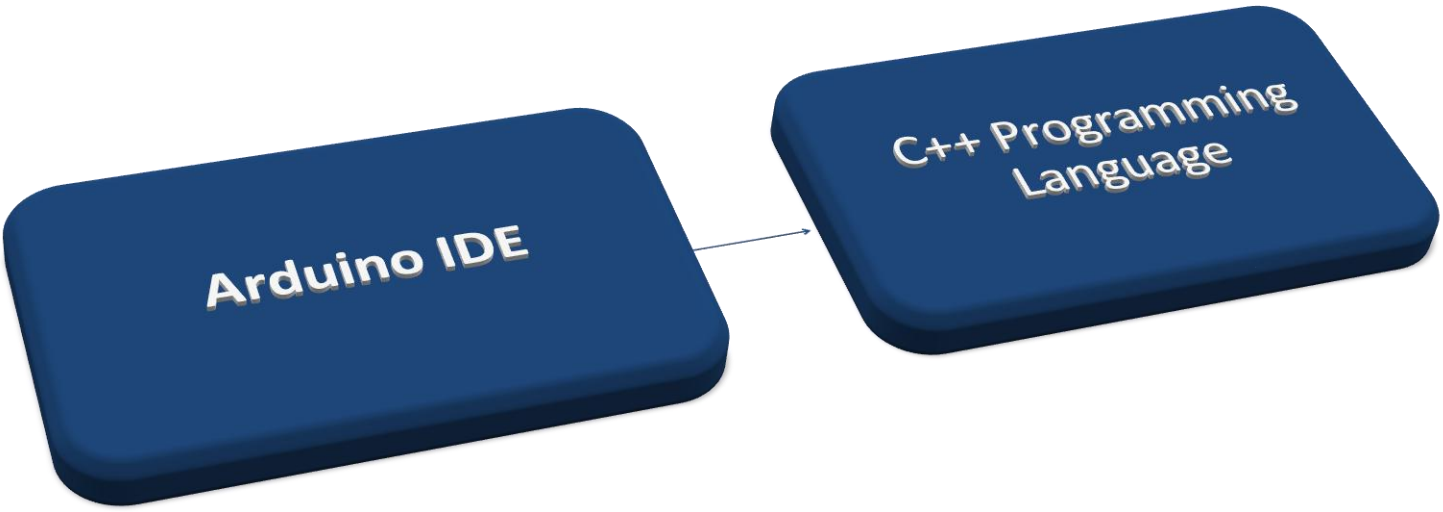


Figure 01 : Arduino UNO R3

Figure 02: Light Dependent Resistor.

Software Components:



- **Arduino IDE** – Used for writing, compiling, and uploading the program to Arduino Uno.
- **C++ Programming Language** – Used for coding the system logic and controlling the behavior of the Arduino-based Smart Automatic Darkness Detection Lighting System.

CIRCUIT DIAGRAM:

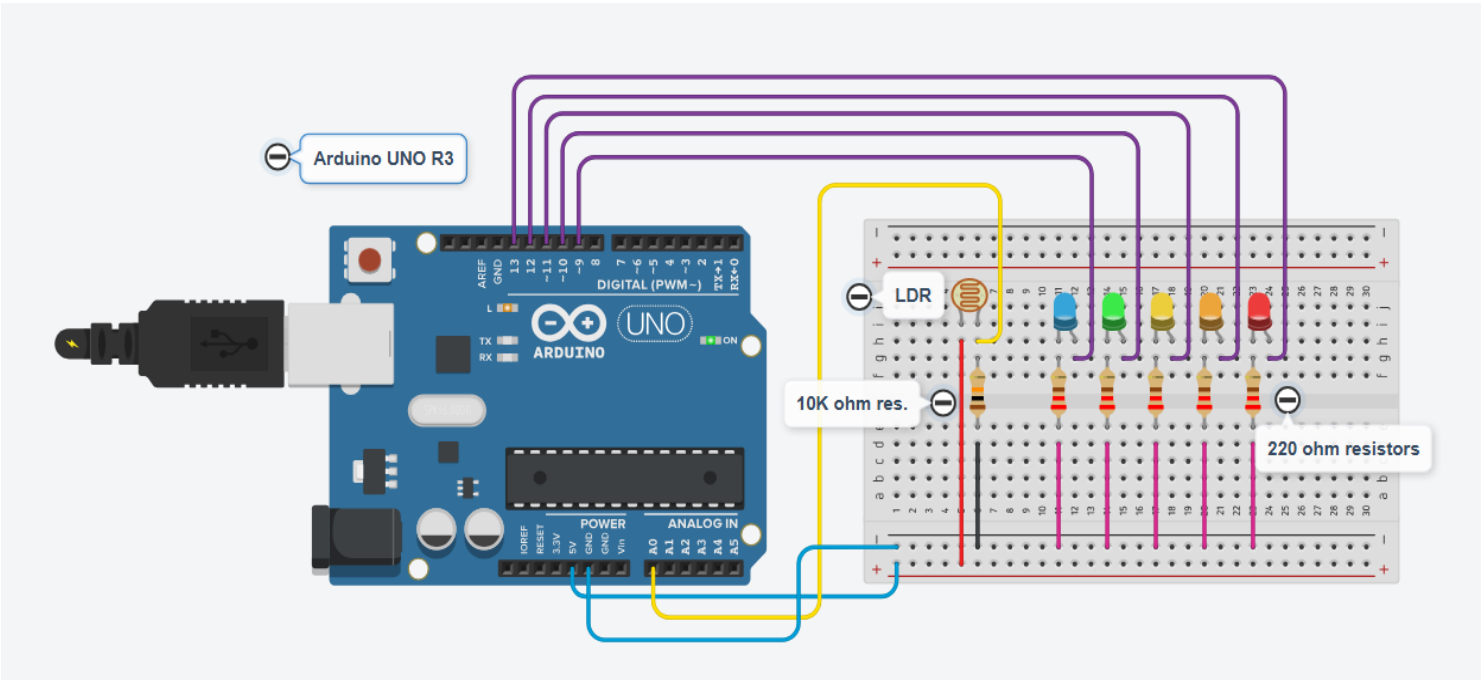


Figure 03 : Circuit Diagram of Automatic Darkness Light Control Project.

EXPECTED OUTCOME :

- The system will automatically control the light based on ambient light conditions
- The LED will turn ON during darkness and turn OFF in sufficient light
- Reduction in unnecessary electricity consumption through automation
- Elimination of manual switching, improving convenience
- Reliable performance using LDR-based light sensing and Arduino control
- Simple and low-cost implementation suitable for practical use

APPLICATIONS :

- Street lighting systems for automatic ON/OFF control based on daylight conditions
- Home lighting automation to improve convenience and energy efficiency
- Garden and outdoor lighting systems for automatic night-time operation
- Parking areas and campus lighting systems to reduce manual control requirements
- Security lighting systems for automatic activation in dark environments
- Rural and remote area lighting solutions where manual operation is difficult
- Energy-efficient smart lighting systems in residential and commercial buildings

ADVANTAGES :

- Reduces electrical energy consumption through automatic control
- Eliminates the need for manual operation of lights
- Simple and low-cost system implementation
- Reliable and efficient performance using LDR-based sensing
- Suitable for various real-life lighting applications

LIMITATIONS :

- Performance depends on proper calibration of the LDR threshold level.
- May be affected by sudden changes in ambient light conditions.
- Limited accuracy in environments with unstable or mixed lighting sources.
- Requires a stable power supply for continuous operation.
- Not suitable for very complex lighting control systems without modification.

FUTURE IMPROVEMENTS :

The system can be further improved by integrating IoT technology, which would allow remote monitoring and control through mobile or web applications. This would enhance flexibility and user convenience by enabling manual override and real-time system management.
A motion sensor can also be added to improve energy efficiency by activating the light only when movement is detected, along with darkness detection. In addition, the system can be powered using solar energy to make it more sustainable and environmentally friendly.
Future versions can also focus on improving sensor accuracy through better calibration techniques and expanding the system for large-scale applications such as smart street lighting networks and smart city infrastructure.

BUDGET ESTIMATION :

Component	Quantity	Estimated Cost (BDT)
Arduino Uno Board	1	600 – 900
LDR Sensor	1	20 – 50
Resistors	2–3	10 – 20
LED	1–2	10 – 20
Breadboard	1	150 – 250
Connecting Wires	Set	50 – 100
Transport Cost	1	300

| Total Estimated Cost | | 1140 – 1640 BDT (approx.) |

CONCLUSION :

This project demonstrates an automatic lighting control system using an LDR and Arduino, which operates based on surrounding light intensity. The system automatically turns the light ON during darkness and OFF when sufficient light is available, ensuring efficient energy usage.

It reduces the need for manual operation and helps in minimizing unnecessary electricity consumption. The design is simple, low-cost, and reliable, making it suitable for real-life applications such as home lighting, street lighting, and other outdoor lighting systems.